Assignment1

1. What is the range of N bits integers with 2’s complement（补码）? What is the range of N bits integers with 1’s complement（反码）? Are they the same? Explain the reasons. (10 points)

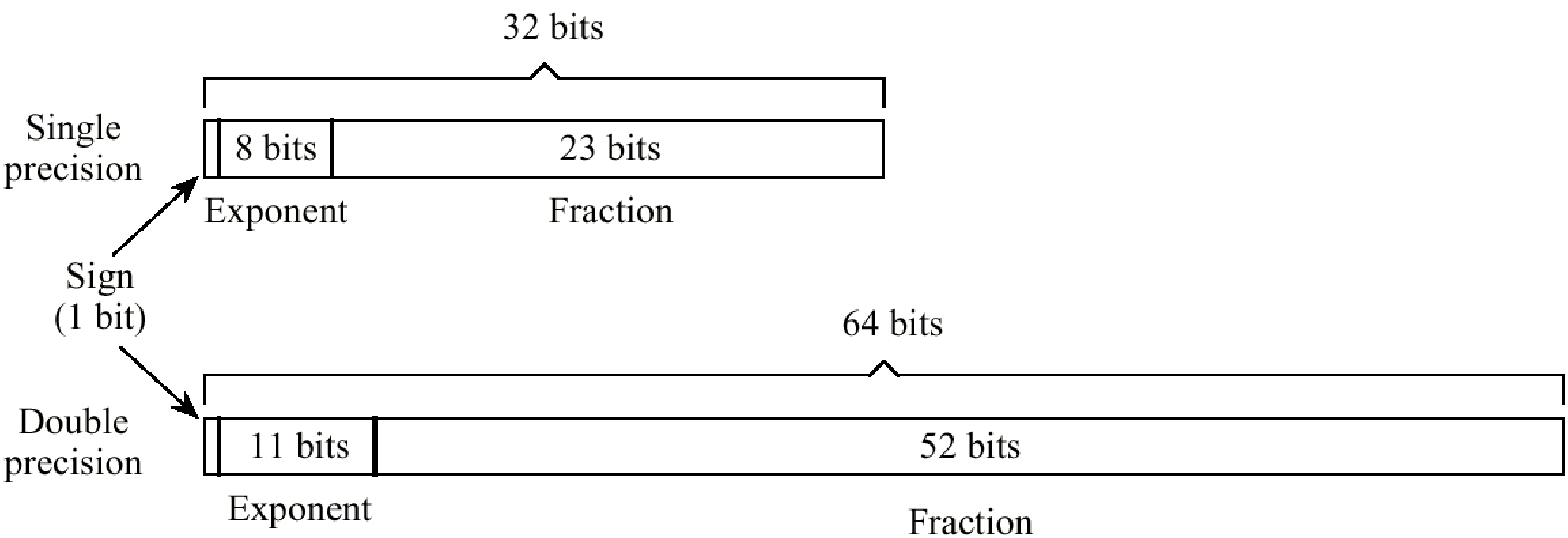
2. Represent the following decimal floating numbers using binary numbers following IEEE754 single precision standard.

$$3\frac{3}{8}$$

$$-2\frac{1}{2}$$

Reference: an IEEE754 single precision number is calculated as:

$$(-1)^{sign\ bit} * (1 + mantissa) * 2^{(\text{exponent}-127)}$$



3. Suppose registers E and F contained $(AA)_{hex}$ and $(CC)_{hex}$, respectively. What bit pattern would be in register D after executing each of the following instructions (see appendix)? .

A. 7DEF ____________

B. 8DEF ____________

C. 9DEF ____________

4. High definition video can be delivered at a rate of 30 frames per second (fps) where each frame has a resolution of 1920 1080 pixels using 3Bytes per pixel. Can an uncompressed video stream of this format be sent over a USB 1.1 serial port? USB 2.0 serial port? USB 3.0 serial port? Why?

(Note: The maximum speeds of USB 1.1, USB 2.0, and USB 3.0 serial ports are 12Mbps, 480Mbps, and 5Gbps respectively.)

5. The following is a routine encoded in the machine language described in the language description table. Explain (in a single sentence) what the routine does. (Explain what the entire routine does as a unit rather than reciting what each instruction does.)

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210F
12A0
8212
32A0
```

6. Suppose the memory cells at addresses 20 through 28 contain the following bit patterns (the program counter contains 20):

Address	Contents
20	12
21	20
22	32
23	30
24	B0
25	21
26	20
27	C0
28	00

(1)What bit patterns will be in register 0, 1 and 2 when the machine halts?

(2)What bit pattern will be in the memory cell at address 30 when the machine halts?

(3) What bit pattern will be in the memory cell at address B0 when the machine halts?

7. Using the machine language described in the appendix, write a sequence of instructions that will place a 1 in the most significant bit of the memory cell at address A0 without disturbing the other bits.

- 8. Suppose the registers 4 and 5 contain the bit patterns 3C and C8, respectively. What bit pattern is left in register 0 after executing each of the following instructions:
- 5045
- 7045
- 8045
- 9045

9. Suppose the memory cells at addresses AF through B1 in the machine described in the Appendix contain the following bit patterns

Address	Contents
AF	B0
B0	B0
B1	AF

What would happen if we start the machine with its program counter containing AF?